

Computer Networking

Introduction

Introduction

- A computer network is defined as an interconnected collection of autonomous computer systems (hardware and Software) that can communicate together.
 - Brought about by the merging of computers and communication systems.

Introduction

- Interconnected means capable of exchanging information
 - Through Wired or wireless connection
- Autonomous means, that there is no dictatorship.

Introduction

- Computer networks should not be confused with Distributed systems.
- Distributed Systems are computer networks with special software that makes the existence of several computers invisible to the user.
 - This special software usually layered on top of the operating systems is known as middleware

Uses of computer networks

Uses of computer networks

- Business applications
- Home network applications
- Mobile network applications

Uses of computer networks

- Additional Reading: Computer Networks by Andrew Tanenbaum Section 1.1

Uses of computer networks- Business applications

Uses of computer networks- Business applications

- Popular uses of computer networks in businesses
 - Resource sharing
 - Communication
 - E-Commerce
 - B2B
 - B2C

Resource sharing

- Makes all programs, data and equipment available to anyone on the network without regard to the physical location of the resource and the user.
 - Resource refers to both **hardware and software** components that can be usefully shared.

Resource sharing

- In general resource sharing provides mechanisms for sharing resources such as:
 - Hardware devices e.g. laser printers, massive archival storage devices and other peripherals
 - Specialized hardware devices(e.g. high speed array processor)
 - Files at remote sites

Resource sharing

- Though there has been a dramatic decrease in cost of micro/minicomputers the cost of peripheral devices has not declined as dramatically.
- The sharing of expensive resources such as high volume laser printers, special purpose processors etc, is cost effective.

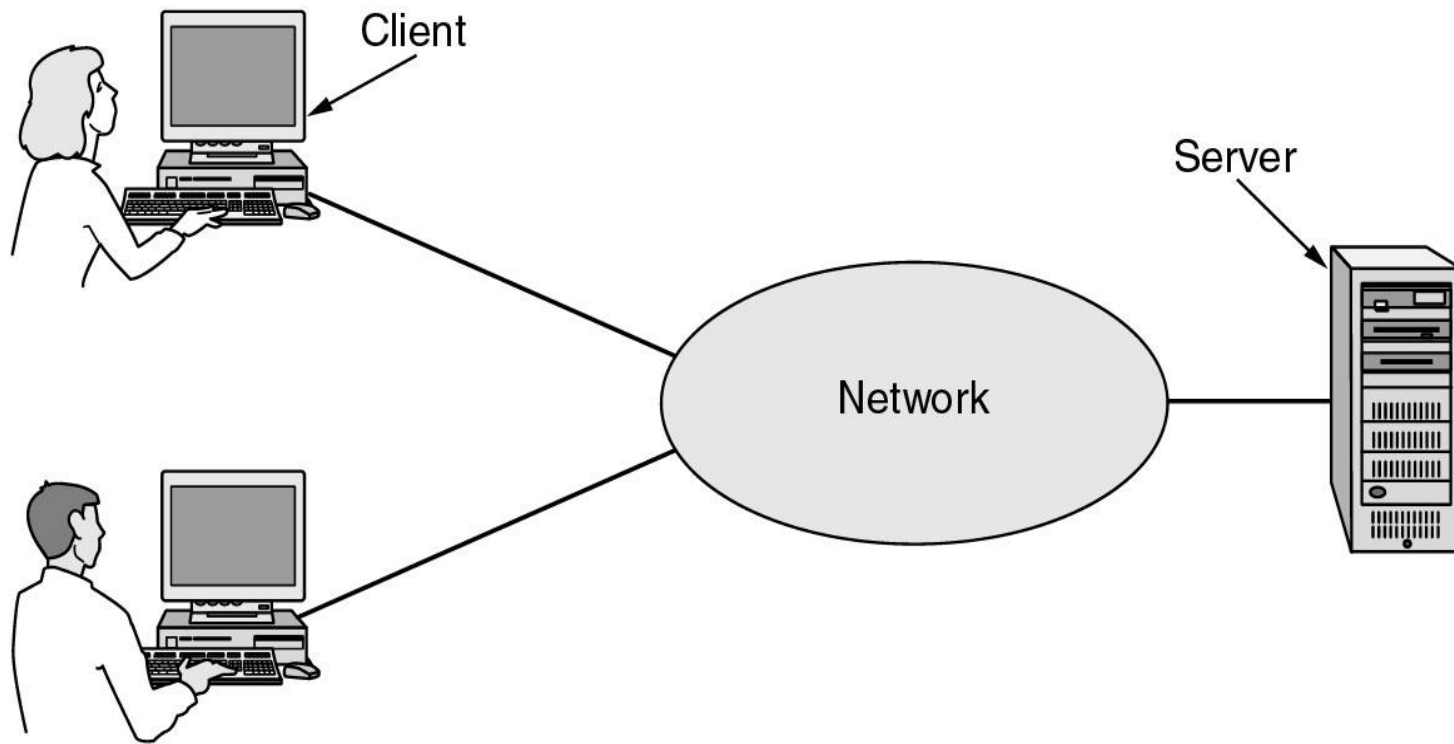
Resource sharing

- Data/Information **sharing** is also important and most likely to occur in a single enterprise.
- Information shared may include:
 - Customer records
 - Inventories
 - Financial statements etc

Resource sharing- client server model

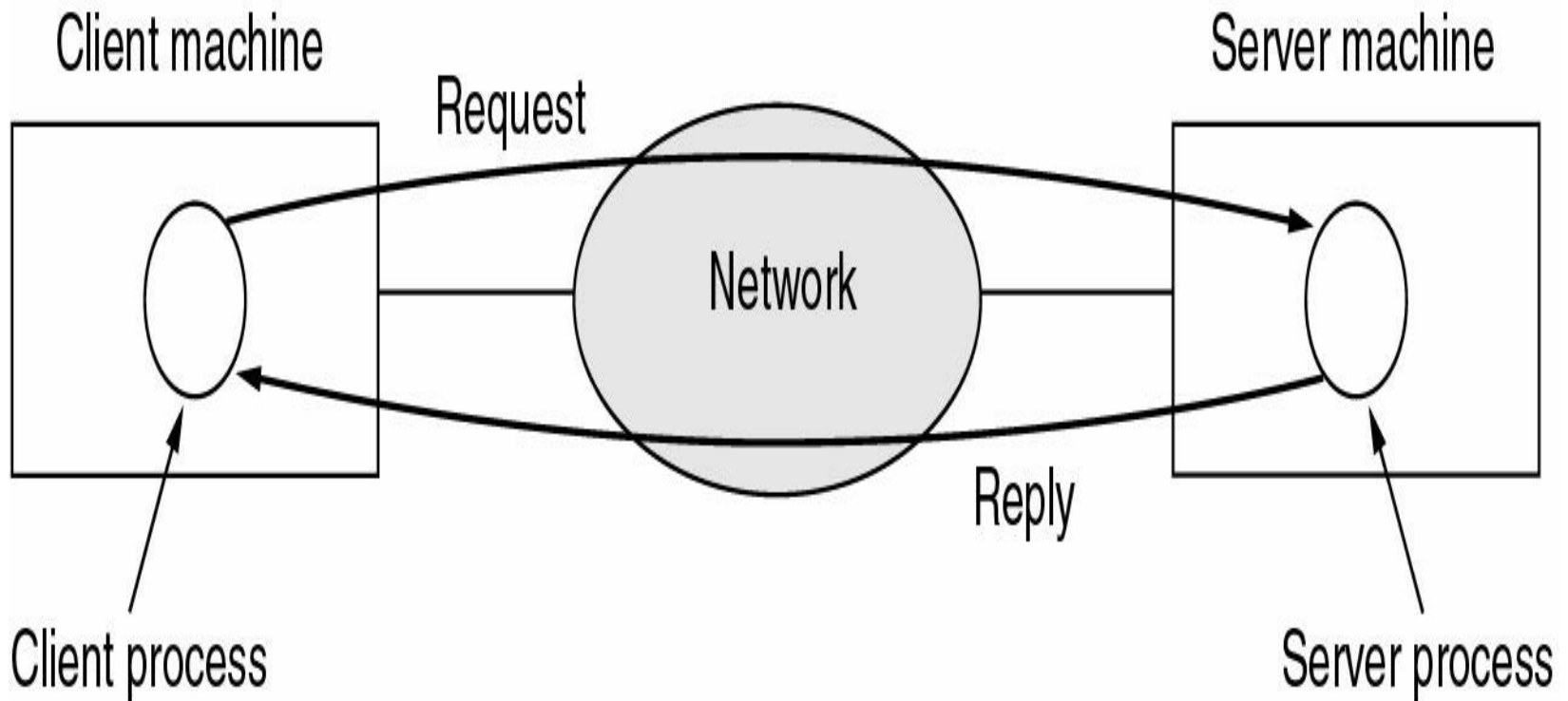
- Data are stored on powerful computers called **servers** – centrally housed and managed by system administrators
- Employees access information through simpler machines on their desks called **clients**

Resource sharing- client server model



A network with two clients and one server

Resource sharing- client server model



Requests and replies in client server model

Communication

- Computer Networks make human-to-human communication over a wide geographic and organizational spread easier.
- Provides a powerful communication medium among employees.

Communication

- Improved communication is made possible via
 - E-mail - Communication by electronic mail, to many users, has become more attractive and virtually replaced traditional paper mail, telephone or even Fax.
 - Shared document writing - when one worker makes a change to an online document, the others can see the change immediately
 - Video conferencing- employees at distant locations can hold a meeting, seeing and hearing each other and writing on a shared virtual blackboard

Electronic commerce - B2B

- Companies can do business with other companies electronically
 - e.g manufacturer can place orders electronically to different suppliers

Electronic commerce- B2C

- Doing business with consumers over the internet
- Many companies provide catalogs of their goods and services online and take orders online. Makes shopping convenient
- Examples: Bookstores, music vendors, Airlines, Hotels.

Uses of computer networks- Home network applications

Home network applications

- Initially for Word processing, games
- Now- more for Internet access

Home network applications-Internet access

Popular uses of the internet for home users

- Access to remote information
- Person-to-person communication
- Interactive entertainment
- Electronic commerce

Home network applications-access to remote information

- Access to remote information comes in many forms
 - Surfing the world wide web for information-science, business, health,sports etc
 - Online newspapers (plus magazines and scientific journals)
 - Digital library e.g. ACM, IEEE

Home network applications-person to person communication

- Communication oriented applications
 - E-mail
 - Instant messaging- allows people to type messages at each other in real time
 - Chat room- multi person version of instant messaging. A message type by a member is seen by everyone in the room
 - Worldwide news groups- Topic oriented discussions
 - Peer to peer communication- individuals form a loose group and can communicate with each other
 - Sharing copyrighted music e.g. Napster
 - Sharing public domain music, photos etc
 - Internet telephone calls, video phone, internet radio
 - Telelearning

Home network applications- interactive entertainment

- Video on demand
- Game playing

Home network applications- Electronic commerce

- Home shopping
- Access to financial institutions- electronic payment of bills
- On-line auctions- consumer to consumer

Forms of E-commerce

Tag	Full name	Example
B2C	Business-to-consumer	Ordering books on-line
B2B	Business-to-business	Car manufacturer ordering tires from supplier
G2C	Government-to-consumer	Government distributing tax forms electronically
C2C	Consumer-to-consumer	Auctioning second-hand products on-line
P2P	Peer-to-peer	File sharing

Mobile users

Mobile users

- Uses of wireless networks
 - Portable office
 - Employer keeping in touch with home
 - Military
 - Emerging wireless mobile applications

Mobile network users

- Portable office
 - People want to be connected to their base (office or home) when away or en-route (land, sea or air)
 - Portable electronic equipment: Notebook computers or personal Digital assistants
 - Tasks users may want to perform: send and receive electronic emails, faxes, telephone calls; surf the web; access remote files; log onto to remote machines
- Wireless networks facilitate computing when away or enroute.

Mobile network users

- Employer keeping in touch with home
 - Fleet of trucks, taxis, delivery vehicle, and repair persons

Mobile network users

- Military
 - Relying on a local networking infrastructure when fighting war is not a good idea.

Mobile users

- Wireless computing vs Mobile computing
 - wireless networking and mobile computing are interrelated but are not the same
 - A notebook may be plugged into a wired network User has mobility but without a wireless network
 - There is a distinction between fixed wireless and mobile wireless
 - Some wireless computers are not mobile

Mobile network users

Wireless	Mobile	Applications
No	No	Desktop computers in offices
No	Yes	A notebook computer used in a hotel room
Yes	No	Networks in older, unwired buildings
Yes	Yes	Portable office; PDA for store inventory

Combinations of wireless networks and mobile computing

Mobile network users

- Emerging wireless mobile applications
 - Portable office
 - Store Inventory
 - Wireless parking meters
 - Wireless vending machines
 - Meter reading (gas, water, electricity)
 - Wireless smoke detectors
 - M-commerce- conduct business over portable wireless devices- banking, shopping etc
 - Location dependent services: maps, weather forecasts
 - Personal area networks and wearable computers – access to secure rooms

Questions?